Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
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Recitation 7 Solutions

Problem 7.1

We revisit the classic medical test problem where a treatment decision must be made based on the results of a diagnostic test. Let random variable $X \in \mathcal{X} \triangleq \{\mathtt{h}, \mathtt{i}\}$ represent whether the patient is healthy ($X = \mathtt{h}$) or infected ($X = \mathtt{i}$). Suppose 1 patient in 100 is infected, i.e., $p_X(\mathtt{i}) = 0.01$. Meanwhile, we have a test that is 90% accurate. We use $Y \in \mathcal{Y} \triangleq \{\mathtt{-}, \mathtt{+}\}$ to represent whether the test result is negative ($Y = \mathtt{-}$) or positive ($Y = \mathtt{+}$), and $p_{Y|X}(\mathtt{-}|\mathtt{h}) = p_{Y|X}(\mathtt{+}|\mathtt{i}) = 0.9$.

(a) Suppose the test result is positive. Determine the probability that the patient is infected.

Solution:

$$\begin{aligned} p_{X|Y}(\mathtt{i}|\mathtt{+}) &= \frac{p_X(\mathtt{i})\, p_{Y|X}(\mathtt{+}|\mathtt{i})}{p_X(\mathtt{i})\, p_{Y|X}(\mathtt{+}|\mathtt{i}) + p_X(\mathtt{h})\, p_{Y|X}(\mathtt{+}|\mathtt{h})} \\ &= \frac{0.01 \cdot 0.9}{0.01 \cdot 0.9 + 0.99 \cdot 0.1} = \frac{1}{12} \approx 0.083. \end{aligned}$$

(b) The following table defines the cost $c(x, \hat{x})$ (in thousands of dollars) of deciding that the patient's state is $\hat{x}$ when it is x.

x	$c(x, \mathtt{h})$	$c(x, \mathtt{i})$
$\mathtt{h}$	0	1
$\mathtt{i}$	20	5

Determine the decision $\hat{x}$ that minimizes the expected cost if the test result is positive.

Solution: The minimum expected cost decision must satisfy

$$\hat{x}(\mathtt{+}) = \arg\min_{a \in \mathcal{X}} \varphi(a),$$

where

$$\begin{aligned} \varphi(a) &= \mathbb{E}[c(X, a)|Y = \mathtt{+}] \\ &= \sum_{x \in \mathcal{X}} c(x, a)\, p_{X|Y}(x|\mathtt{+}) \\ &= c(\mathtt{h}, a)\, p_{X|Y}(\mathtt{h}|\mathtt{+}) + c(\mathtt{i}, a)\, p_{X|Y}(\mathtt{i}|\mathtt{+}). \end{aligned} \tag{1}$$

We substitute the values of cost $c(\cdot,\cdot)$ from the table above and the values of the posterior distribution $p_{X|Y}(\cdot|\cdot)$ from part (a) into (1) to obtain

$$\varphi(\mathtt{h}) = c(\mathtt{h},\mathtt{h})\, p_{X|Y}(\mathtt{h}|\mathtt{+}) + c(\mathtt{i},\mathtt{h})\, p_{X|Y}(\mathtt{i}|\mathtt{+}) = 0 \cdot \frac{11}{12} + 20 \cdot \frac{1}{12} = \frac{5}{3},$$
$$\varphi(\mathtt{i}) = c(\mathtt{h},\mathtt{i})\, p_{X|Y}(\mathtt{h}|\mathtt{+}) + c(\mathtt{i},\mathtt{i})\, p_{X|Y}(\mathtt{i}|\mathtt{+}) = 1 \cdot \frac{11}{12} + 5 \cdot \frac{1}{12} = \frac{4}{3}.$$

Since $\varphi(\mathtt{i}) < \varphi(\mathtt{h})$, $\hat{x}(\mathtt{+}) = \mathtt{i}$ minimizes the expected cost. In this case, the cost of deciding the patient is healthy when they are infected is so high that it causes us to decide that the patient is infected even if the probability of infection is only about 8%.

(c) Now suppose $c(x, \hat{x}) = \mathbb{1}(x \neq \hat{x})$. How does that affect you decision?

Solution: Recall that the cost in this part corresponds to the minimum probability of error (MPE) or, equivalently, maximum *a posteriori* (MAP) decisions. Since $p_{X|Y}(\mathtt{h}|\mathtt{+}) > p_{X|Y}(\mathtt{i}|\mathtt{+})$, we have $\hat{x}(\mathtt{+}) = \mathtt{h}$.

To see how this solution follows from the derivation of the expected cost, we substitute the new cost values into (1) to obtain

$$\varphi(\mathtt{h}) = c(\mathtt{h},\mathtt{h})\, p_{X|Y}(\mathtt{h}|\mathtt{+}) + c(\mathtt{i},\mathtt{h})\, p_{X|Y}(\mathtt{i}|\mathtt{+}) = p_{X|Y}(\mathtt{i}|\mathtt{+}) = \frac{1}{12},$$
$$\varphi(\mathtt{i}) = c(\mathtt{h},\mathtt{i})\, p_{X|Y}(\mathtt{h}|\mathtt{+}) + c(\mathtt{i},\mathtt{i})\, p_{X|Y}(\mathtt{i}|\mathtt{+}) = p_{X|Y}(\mathtt{h}|\mathtt{+}) = \frac{11}{12}.$$

Now $\varphi(\mathtt{i}) > \varphi(\mathtt{h})$, and $\hat{x}(\mathtt{+}) = \mathtt{h}$ minimizes the expected cost.

Now suppose we run the test n times, producing test results $Y_1, \ldots, Y_n \in \mathcal{Y}$. Assume that $Y_1, \ldots, Y_n$ are independent and identically distributed given the patient's state X, i.e.,

$$p_{Y_1,\ldots,Y_n|X}(y_1, \ldots, y_n|x) \triangleq \prod_{i=1}^{n} p_{Y|X}(y_i|x), \quad \text{for all } x \in \mathcal{X},\ y_1, \ldots, y_n \in \mathcal{Y}.$$

We define $k \triangleq \sum_{i=1}^{n} \mathbb{1}(y_i = \mathtt{+})$ to be the number of positive test results.

(d) Determine the maximum likelihood (ML) decision $\hat{x}_{\mathrm{ML}}(y_1, \ldots, y_n)$ given the observations $Y_1 = y_1, \ldots, Y_n = y_n$, expressing your answer in terms of k and n.

Solution: The ML estimate is defined as follows:

$$\hat{x}_{\mathrm{ML}}(y_1, \ldots, y_n) = \arg\max_{x \in \mathcal{X}} p_{Y_1,\ldots,Y_n|X}(y_1, \ldots, y_n|x).$$

We compute

$$p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{h}) = \prod_{i=1}^{n} p_{Y|X}(y_i|\mathtt{h}) = 0.9^{n-k}\cdot 0.1^{k} = \frac{9^{n-k}}{10^n}, \tag{2}$$

$$p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{i}) = \prod_{i=1}^{n} p_{Y|X}(y_i|\mathtt{i}) = 0.9^{k}\cdot 0.1^{n-k} = \frac{9^{k}}{10^n}. \tag{3}$$

and assign $\hat{x}_{\mathrm{ML}}(y_1,\ldots,y_n) = \mathtt{i}$ if

$$p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{i}) > p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{h})$$

which after substituting (2) and (3) corresponds to

$$\frac{9^k}{10^n} > \frac{9^{n-k}}{10^n},$$

i.e., $k > n/2$. Thus

$$\hat{x}_{\mathrm{ML}}(y_1,\ldots,y_n) = \begin{cases} \mathtt{i} & k > n/2 \\ \mathtt{h} & \text{otherwise.} \end{cases}$$

i.e., we decide the patient is infected if more than half of the test results are positive and decide they are healthy if fewer than half of the test results are positive. What if the number of positive results is exactly half of all tests? In this case, the likelihood of the observed data is the same whether the patient is healthy or infected, which means that the ML rule is a tie and both solutions maximize the likelihood of the data.

(e) Determine the maximum *a posteriori* (MAP) decision $\hat{x}_{\mathrm{MAP}}(y_1,\ldots,y_n)$ given the observations $Y_1 = y_1,\ldots,Y_n = y_n$, expressing your answer in terms of k and n.

Solution: The MAP decision rule is as follows:

$$\begin{aligned}\hat{x}_{\mathrm{MAP}}(y_1,\ldots,y_n) &= \arg\max_{x\in\mathcal{X}} p_{X|Y_1,\ldots,Y_n}(x|y_1,\ldots,y_n) \\ &= \arg\max_{x\in\mathcal{X}} p_X(x)\, p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|x).\end{aligned}$$

We decide $\hat{x}_{\mathrm{MAP}}(y_1,\ldots,y_n) = \mathtt{i}$ if

$$\begin{aligned} p_X(\mathtt{i})\, p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{i}) &> p_X(\mathtt{h})\, p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{h}), \\ 0.01\cdot 9^k &> 0.99\cdot 9^{n-k}, \\ k &> \frac{n}{2} + \frac{\log(99)}{2\,\log(9)} \approx \frac{n}{2} + 1.046, \end{aligned}$$

where we used (2) and (3) to obtain the second inequality. Hence

$$\hat{x}_{\text{MAP}}(y_1, \ldots, y_n) = \begin{cases} \mathtt{i} & k > n/2 + 1.046 \\ \mathtt{h} & \text{otherwise.} \end{cases}$$

Note that as $n \to \infty$, the MAP decision rule in this part is almost the same as that of the ML rule we computed in part (d). In fact, this is true in general: when there is a lot of data, the MAP decision gets very close to the ML decision.

What does this mean? The MAP decision is just the ML decision with the effect of the prior beliefs incorporated in it as a multiplicative factor. The ML decision is purely driven by the data, and once we incorporate the prior it becomes the MAP decision. When there is a lot of data, it doesn't really matter what we thought *a priori*, the data will dictate the decision anyway. In other words, there's no data like lots of data!

(f) Determine the decision rule that minimizes the expected cost defined in part (b) based on the observations $Y_1 = y_1, \ldots, Y_n = y_n$, expressing your answer in terms of k and n.

Solution: As in the solution to part (b), we choose the decision that minimizes the expected cost. Specifically, we decide $\hat{x}(y_1, \ldots, y_n) = \mathtt{i}$ if

$$\varphi(\mathtt{h}) > \varphi(\mathtt{i}). \tag{4}$$

Using that

$$\begin{aligned} \varphi(\mathtt{h}) &= c(\mathtt{h},\mathtt{h})\, p_{X|Y_1,\ldots,Y_n}(\mathtt{h}|y_1,\ldots,y_n) + c(\mathtt{i},\mathtt{h})\, p_{X|Y_1,\ldots,Y_n}(\mathtt{i}|y_1,\ldots,y_n) \\ &\propto c(\mathtt{h},\mathtt{h})\, p_X(\mathtt{h})\, p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{h}) + c(\mathtt{i},\mathtt{h})\, p_X(\mathtt{i})\, p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{i}) \end{aligned}$$

and

$$\begin{aligned} \varphi(\mathtt{i}) &= c(\mathtt{h},\mathtt{i})\, p_{X|Y_1,\ldots,Y_n}(\mathtt{h}|y_1,\ldots,y_n) + c(\mathtt{i},\mathtt{i})\, p_{X|Y_1,\ldots,Y_n}(\mathtt{i}|y_1,\ldots,y_n) \\ &\propto c(\mathtt{h},\mathtt{i})\, p_X(\mathtt{h})\, p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{h}) + c(\mathtt{i},\mathtt{i})\, p_X(\mathtt{i})\, p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{i}), \end{aligned}$$

the condition (4) corresponds to, after rearranging terms,

$$\begin{aligned} &\big[c(\mathtt{i},\mathtt{h}) - c(\mathtt{i},\mathtt{i})\big] p_X(\mathtt{i})\, p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{i}) \\ &\qquad > \big[c(\mathtt{h},\mathtt{i}) - c(\mathtt{h},\mathtt{h})\big] p_X(\mathtt{h})\, p_{Y_1,\ldots,Y_n|X}(y_1,\ldots,y_n|\mathtt{h}). \end{aligned}$$

After substituting values, the condition is

$$15 \cdot 0.01 \cdot 9^k > 1 \cdot 0.99 \cdot 9^{n-k},$$

i.e.,

$$> \frac{n}{2} + \frac{\log \frac{1 \cdot 0.99}{15 \cdot 0.01}}{2 \log(9)} \approx \frac{n}{2} + 0.43,$$

so

$$\hat{x}(y_1, \ldots, y_n) = \begin{cases} \texttt{i} & k > n/2 + 0.43 \\ \texttt{h} & \text{otherwise.} \end{cases}$$

Once again, we see something perhaps surprising—if n is large, then the decision rule in this part is very nearly the same as the ML solution in part (d). This is interesting – the prior *and* the cost are essentially irrelevant! Thus, data swamps not only the prior, but also the cost. With enough data, it doesn't really matter what our prior beliefs are, and what the cost of each decision is—we will end up going where the data tells us. This points to the great value of the ML decision—ultimately, with lots of data, the other decisions that we defined reduce to the ML decision rule. We will see the ML criterion again quite soon.